Log Analysis:

Log is an event or an action that is happened.

The purpose of log is to give us information about the event – what has happened, when it happened, why it happened and how it happened.

e.g., Security log will tell you who logged in into OS, whether the login was successful or not and the time when it happened.

All we need is Windows operating system to understand logs. There is a tool in windows called **Event Viewer** through which we can see logs. It's where all logs are stored.

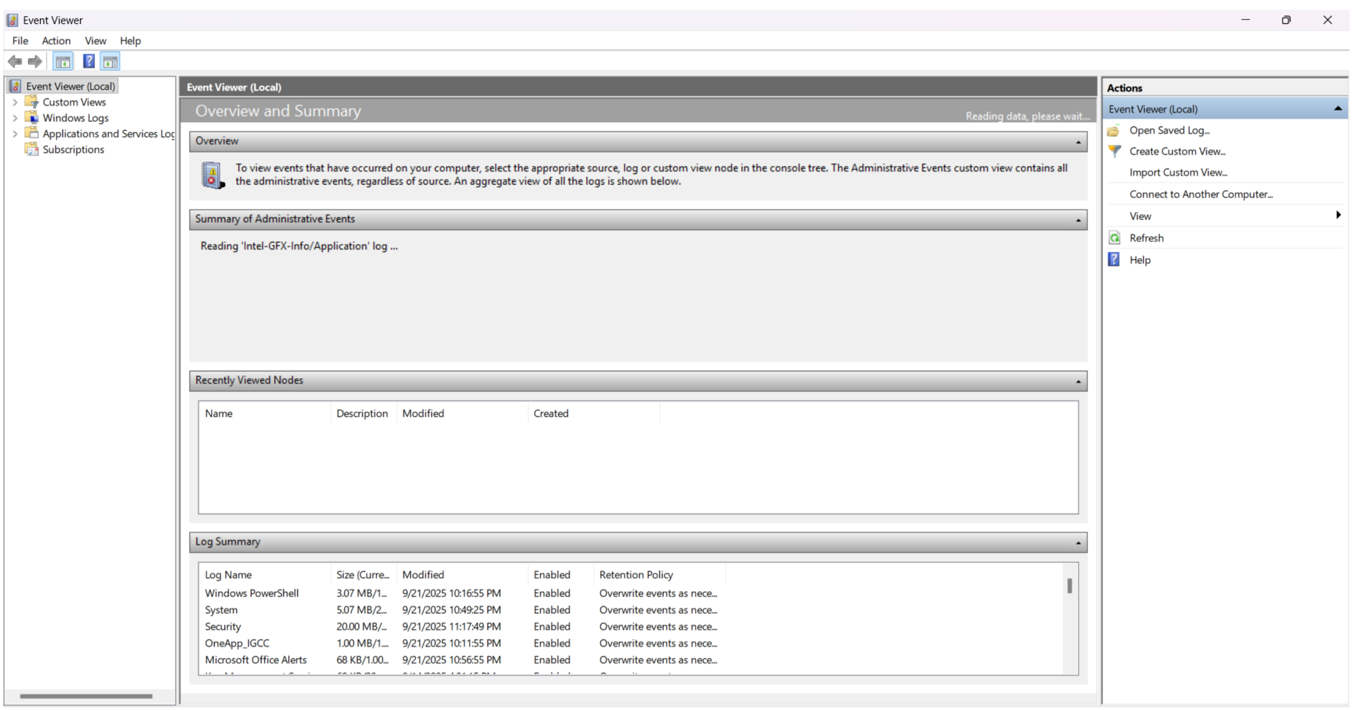
Ways to open Event Viewer:

Search **event viewer** in start menu

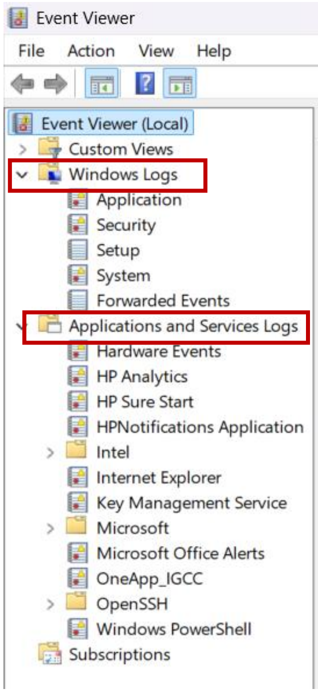
OR

Open **cmd** > Type **eventvwr** (same command works in Run bar)

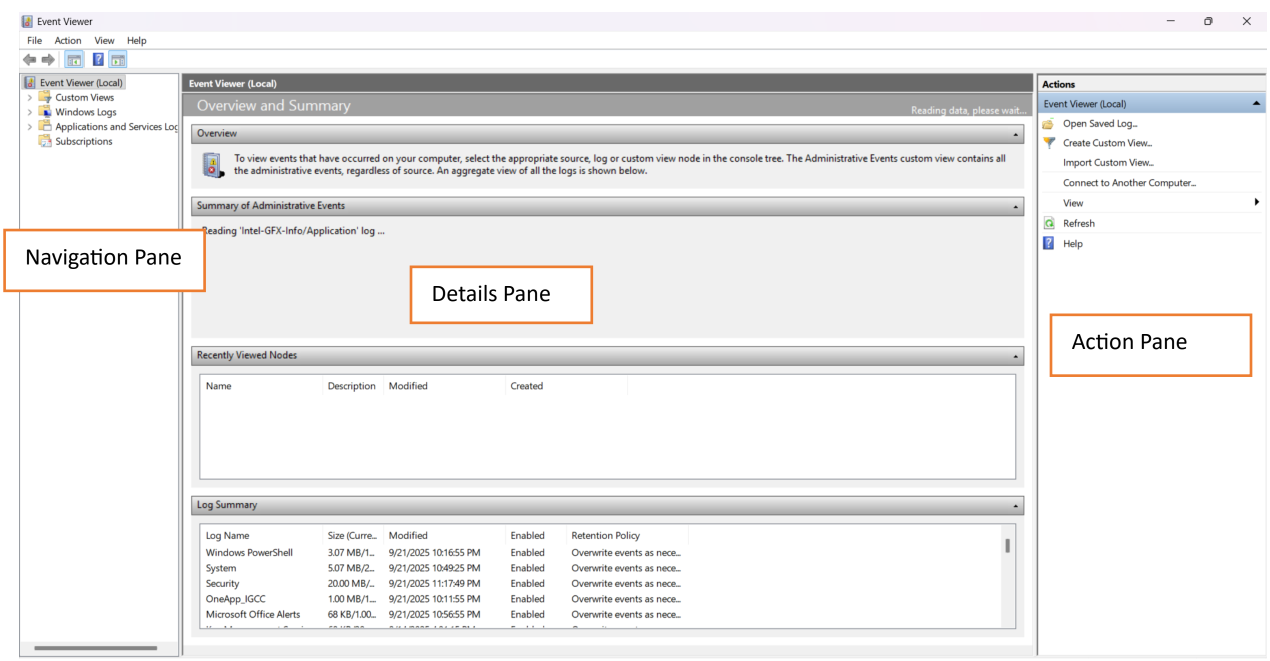
Suppose our windows get corrupted and we don't have GUI access. So, it's better to know how to open it using command prompt.



We have **Windows Log**, **Application and Service Logs**



- The left box is known as **Navigation pane**
- The middle box in Event Viewer is called **Details pane** where all the details are seen.
- The right box is called **Action pane**



Logs Categories:

Application: Records problems or messages from apps you use (like Chrome, Notepad, or Office etc)

System: Shows issues or updates related to Windows itself (like starting/stopping services).

Suppose we download **Splunk** – when we start is, its log will be saved as service was started/closed.

Security: Tracks logins, logouts, and security-related actions (like failed passwords or new users).

Forwarded Events: Collects logs from other computers if one is set up to receive them.

Usually in an organization, one server is set up to receive logs from all other computers on the network. All these logs will come in forwarded events.

Setup: Keeps tracks of things that happen when Windows is being installed or updated.

PowerShell Logs: Shows what scripts or commands were run using PowerShell (good for spotting suspicious activity).

These are important because attackers normally compromise powershell to execute malicious commands.

Sysmon (if enabled): Gives detailed behind-the-scenes info like program launches, file changes or internet access – like CCTV for your computer.

It is a detailed version of your logs. Sysmon is not enabled by default. We have to enable it ourselves.

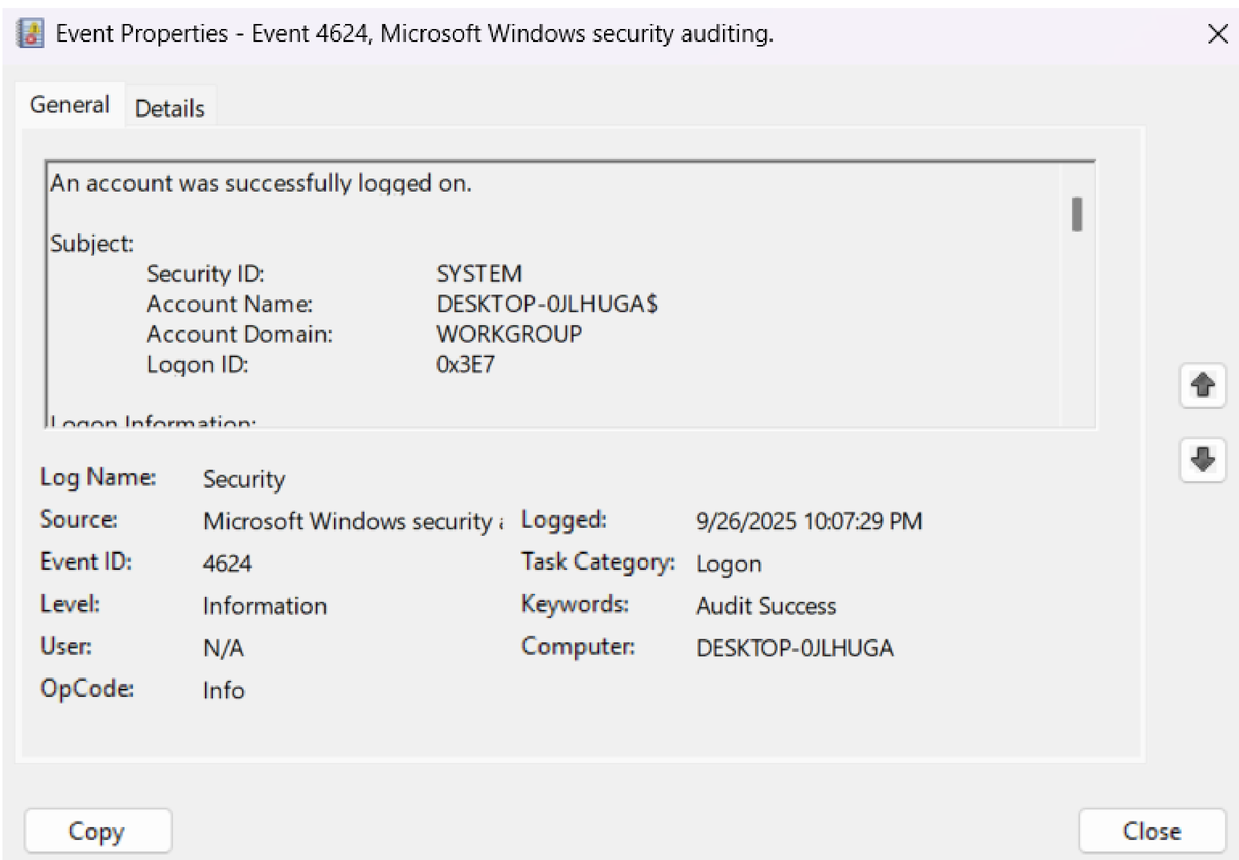
All the events happened in windows have **Event_id** such as login success, login failure, user create or delete. All of these events have specific event-id.

e.g., 4624 is event-id for successful login

Event IDs

Event ID	Event Name	Explanation
4624	Successful Logon	Someone logged into the system successfully.
4625	Failed Logon	Someone tried to log in, but failed (wrong password etc).
4634	Logoff	A user logged off from the system.
4672	Admin Privileges Assigned	A user got admin-level permissions (very sensitive).
4688	Process Created	A new program or script was run.
4689	Process Ended	A program or process was closed.
4697	Service Installed	A new Windows service was added (often used by malware).
4720	User Created	A new user account was created.
4722	User Enabled	A user account was re-enabled (after being disabled).
4723	Password Change Attempt	A user tried to change their password.
4725	User Disabled	A user account was disabled.
4726	User Deleted	A user account was deleted.
4728	Added to Security Group	A user was added to a privileged group like Administrators.
4771	Kerberos Pre-auth Failure	Possible brute force attempt using Kerberos.
5140	Shared File Access	Someone accessed a shared folder over the network.
5156	Allowed Network Connection	A program was allowed to connect to another machine.

Let’s open the detail of a random event.



User is showing N/A because it was not the event triggered by the user. It can be an event which is triggered by a process running in a background.

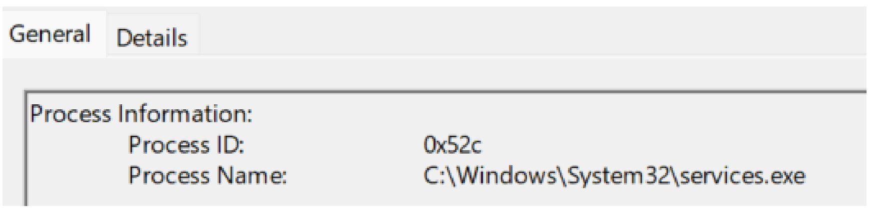
If I manually log in and out of my operating system, I can see my user there.

OpCode is similar to the Level.

Logged is timestamp (on which date and time it is happened)

Keywords (Audit Success means the event is happened successfully)

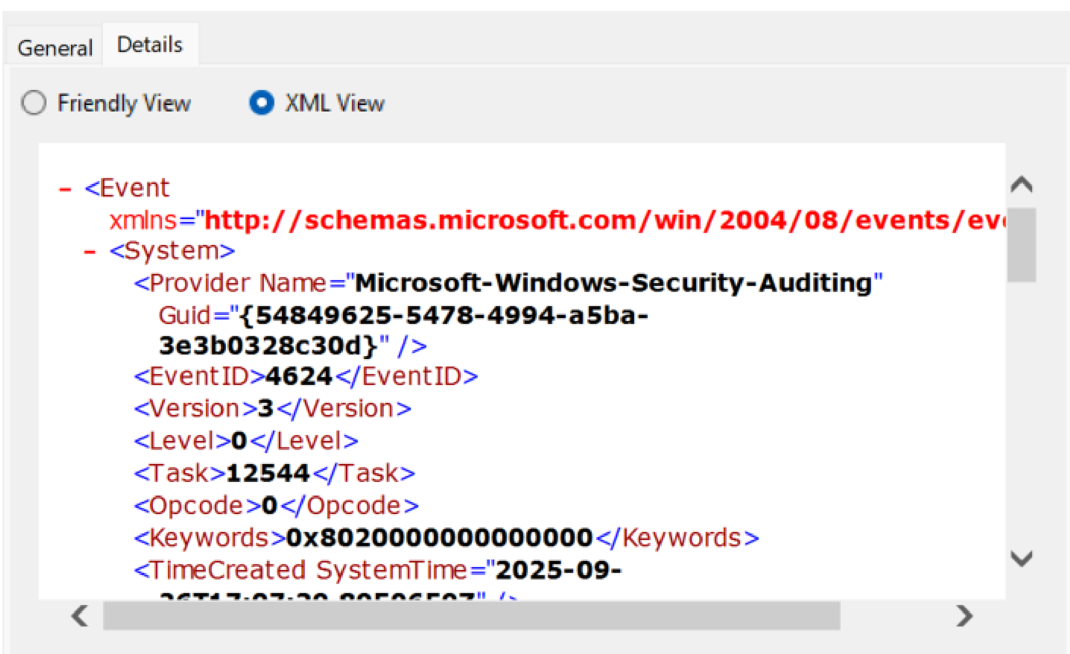
If we scroll down, we can see more details but that is not necessary. The detail which is important is **Process Information**



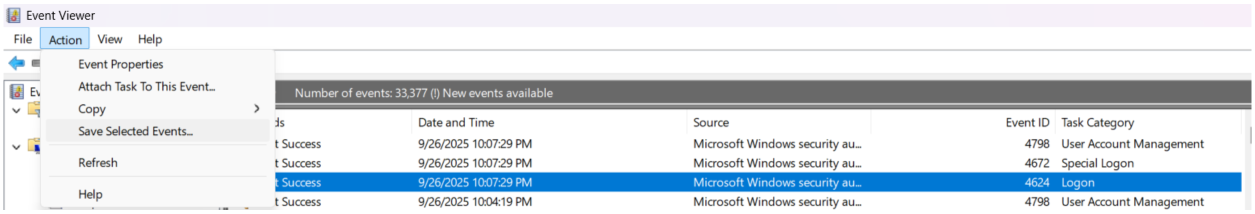
Process name is basically what triggered this event.

If we click on the **Details** tab, we can XML view which as a SOC Analyst is important to me. The reason is when we are exporting Windows logs to any SIEM tool, the backend mechanism is XML. Also, if we enable Sysmon, its entire configuration is based on XML.

We can see event-id here and then we can tell what happened. This is why event-ids are important. It is also important for interview because interviewer might show you this and ask what is happening here. So, by looking at event-id, you can tell successful logon.



We can save the specific event. Click on the event > Click on actions tab > Save

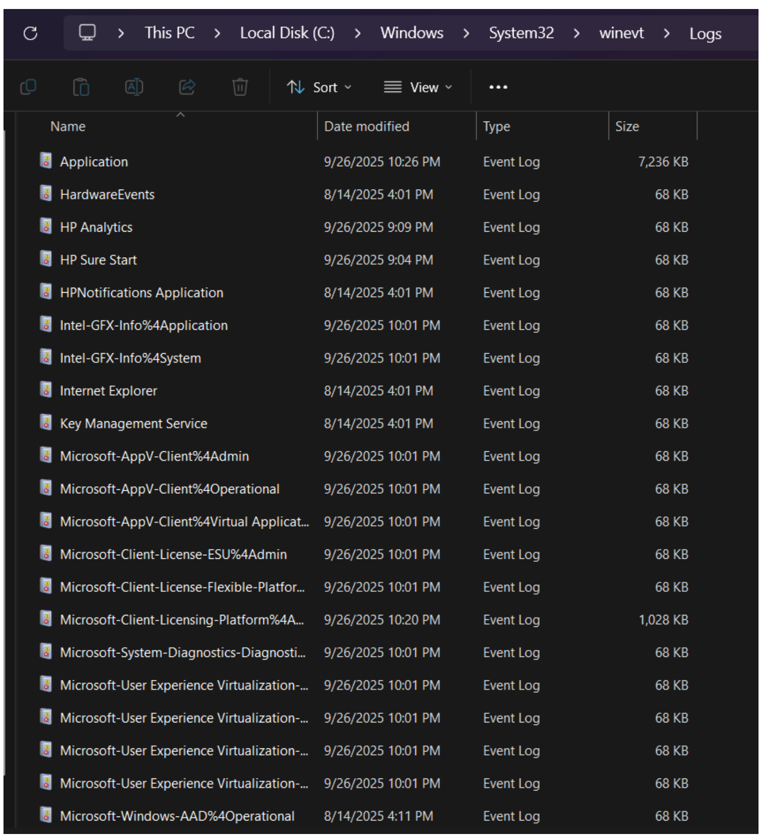


Extension of Windows event file is **.evtx**

We can save events file in these formats:

We can find these logs navigation pane or all about this EventViewer in this path:

C:/ > Windows > System32 > winevt > logs



On the EventViewer, if we click on the **System**, we can see like this

System Number of events: 9,637

Level	Date and Time	Source	Event ID	Task Category
Information	9/26/2025 10:36:10 PM	WindowsUpdateClient	19	Windows Update Agent
Information	9/26/2025 10:18:57 PM	Service Control Manager	7040	None
Information	9/26/2025 10:16:45 PM	Service Control Manager	7040	None
Information	9/26/2025 9:59:35 PM	Kernel-General	16	None
Information	9/26/2025 9:50:50 PM	Kernel-General	16	None
Warning	9/26/2025 9:34:10 PM	hcmon	0	None
Warning	9/26/2025 9:34:10 PM	hcmon	0	None
Warning	9/26/2025 9:15:46 PM	hcmon	0	None
Warning	9/26/2025 9:15:46 PM	hcmon	0	None
Information	9/26/2025 9:10:09 PM	OneCore-DeviceAssociationS...	2500	(7000)
Warning	9/26/2025 9:10:09 PM	hcmon	0	None
Warning	9/26/2025 9:10:09 PM	hcmon	0	None
Warning	9/26/2025 9:10:03 PM	hcmon	0	None
Warning	9/26/2025 9:10:03 PM	hcmon	0	None
Warning	9/26/2025 9:10:00 PM	DistributedCOM	10016	None
Warning	9/26/2025 9:10:00 PM	DistributedCOM	10016	None
Warning	9/26/2025 9:10:00 PM	DistributedCOM	10016	None
Warning	9/26/2025 9:10:00 PM	DistributedCOM	10016	None
Information	9/26/2025 9:09:59 PM	BTHUSB	18	None
Information	9/26/2025 9:09:26 PM	Kernel-Power	105	(100)

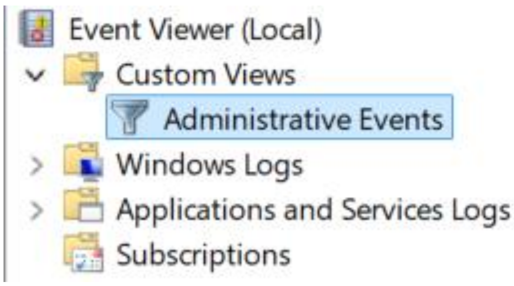
What is this? This is where **Level IDs** come in. Each event has its own level id in Windows. It shows the severity of the event.

Level IDs

LEVEL ID	DESCRIPTION
Critical (30)	Something crashed or seriously broke – fix it right now!
Error (40)	Something went wrong, but it's not urgent (yet).
Warning (50)	Something might go wrong soon – keep an eye on it.
Information (80)	Everything is working fine – just keeping you updated.
Debug (100)	Detailed updates for nerds – not needed unless you're troubleshooting

ID **60**, **70** and **90** are reserved for software developers to use.

Windows has its own **Custom Views** called **Administrative Events**.



It's like you can create your own dashboard like which specific category logs you want to see. Like if I want to view the event logon success only, it will create a custom view for that log only.

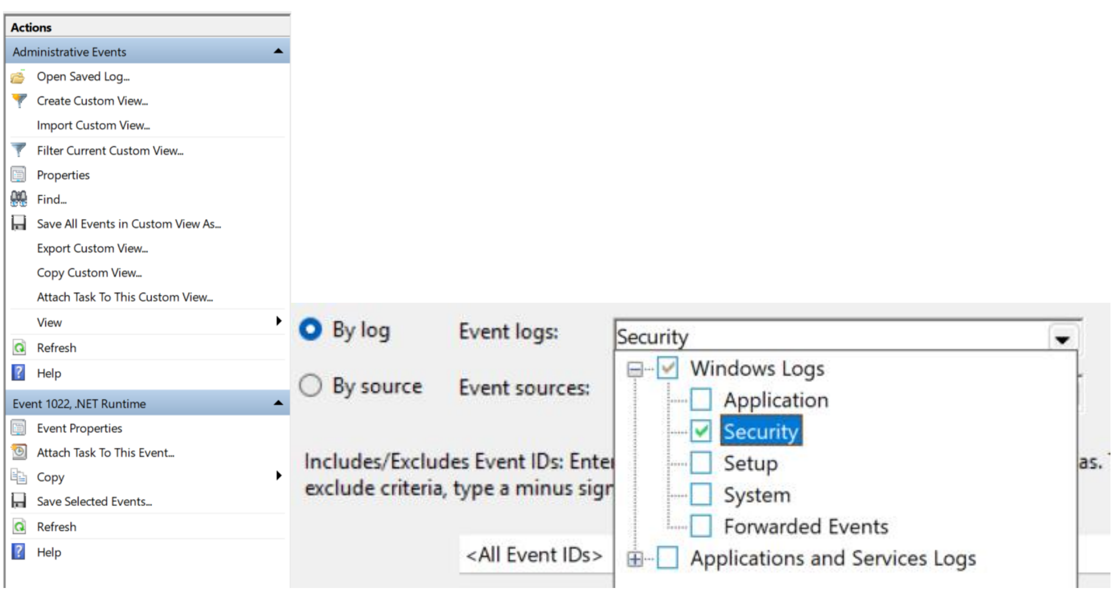
Benefits of Custom Views:

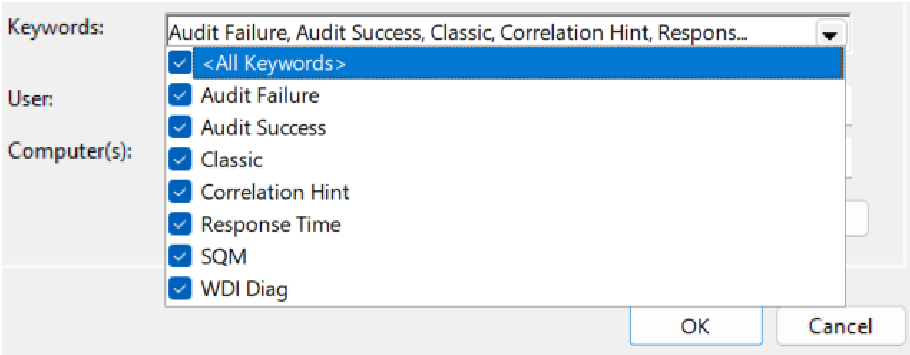
- Easy to understand
- Rather than forwarding all of the logs, companies can create specific custom views and forward that to SIEM tool

Administrative Events tab is a combination of all 5 tabs of Windows Log. But it will show only the view of Error, Critical and Warning events.

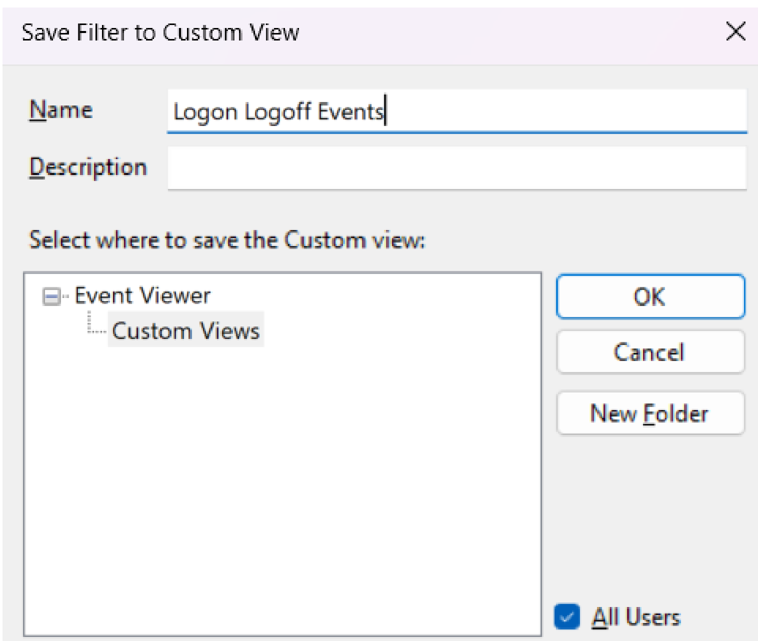
How to create your own Custom View?

Click on Create Custom View > Select the level of Events you want there > Select Event log types which you want to see > If you want all event ids or only specific (if you want specific events to be shown there write their ids separating with comma or you can define a range like 4264-4625 > Select keywords





Click OK



Give it a name > OK

Sysmon | System Monitor:

- Windows logs are like the regular CCTV, they show who came in and out. But attackers are smart they sneak in and hide their actions.
- Sysmon is like a high-definition security camera, it shows what program ran, what files were created, what network connection was made, every detail that regular Windows logs miss.

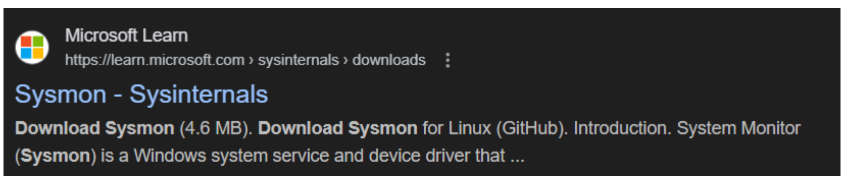
- Sysmon is powerful, but it can be noisy if we don't tell it what to watch and what to ignore. That's where the config file comes in.
- Think of it like setting up your CCTV rules you can say: only record people wearing red, ignore cats, focus on the door not the window.
- We do the same here, we tell Sysmon - watch for PowerShell, watch for rundll32, ignore Notepad or another legitimate/known Program.

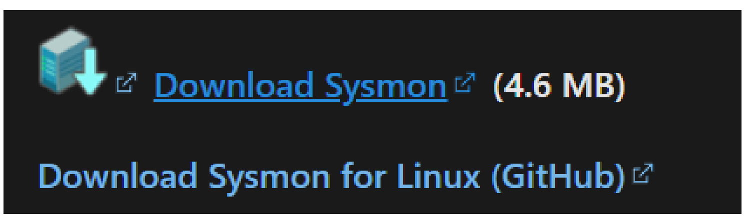
It is not enabled by default. We can download it or enable it.

All of the Sysmon rules are in XML format.

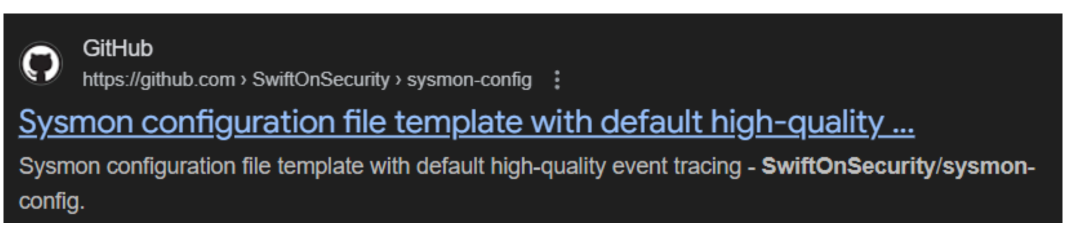
How to enable Sysmon?

Go to the web > Search Sysmon download > Download from the official Microsoft site





There is a file on **GitHub** which have pre-defined rules for Sysmon in XML format. We can download that too if we don't want to make our own file in XML.



After unzipping the file, paste the rule file in the Sysmon directory which we've downloaded from Microsoft site, so we don't have to give the path.

Syson Event IDs:

Sysmon IDs

Sysmon ID	Event Name	Simple Explanation
1	Process Creation	A program or command was executed (including full command line).
3	Network Connection	A program made a connection to another computer or IP.
7	Image Loaded	A DLL or EXE file was loaded into memory.
8	CreateRemoteThread	A process tried to run code inside another process (often used in malware).
10	Process Access	One program accessed another process (like memory scraping).
11	File Created	A file was created on disk.
12	Registry Object Created	A new registry key was added.
13	Registry Value Set	A registry value was changed.
14	Registry Key Renamed	A registry key was renamed.
22	DNS Query	A program asked Windows to resolve a domain name (e.g., google.com).
23	File Delete Detected	A file was deleted.
25	Pipe Connected	A program connected to an inter-process communication channel

You have to remember Sysmon Event ids. **It's very important** in terms of interview and in general as well. If you know the id's, you can configure Sysmon file accordingly and as per your need.

Sysmon is the goldmine in Threat Hunting.

To set up or install Sysmon:

- Run cmd as an administrator
- Follow the guide given on GitHub (from where you download config file)

If you open Event Viewer, under Application and Services Logs > Windows, you can find Sysmon logs because Sysmon itself is an application and process.